Software Requirements Specification (SRS)

Ward Connect

Group No: 10

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1 Introduction

1.1 Purpose

The purpose of this project is to design and develop **Ward Connect**, a ward-level service platform that brings together governance, community services, healthcare, and utilities into a single digital hub. Citizens, ward officials, Kudumbashree groups, and service providers can interact through this application to improve efficiency, accountability, and accessibility of services.

1.2 Definitions and Acronyms

- **Kudumbashree** – Kerala state women empowerment and poverty eradication initiative.
- **AI** – Artificial Intelligence (used for speech-to-text, summarization, chatbot, etc.).
- **DB** – Database PostgreSQL
- **API** – Application Programming Interface (used for integrations).
- **GIS** – Geographic Information System (for location-based tracking).

2 Overall Description

The system integrates a wide variety of ward-level services into a single app. At present, most of these services exist in fragmented or manual form. Ward Connect bridges this gap by consolidating:

- Kudumbashree services (meetings, loans, attendance).
- Waste management services (pickup, complaints).
- Vehicle booking services.
- Education and job support services.
- Utility and billing services.
- Civic issue reporting.
- Healthcare support.
- Local shop services.

2.1 User Needs

- **Citizens**: Access all essential ward services in one app.
- **Ward Officials/Admins**: Manage schedules, complaints, waste collection, and generate analytics.
- **Kudumbashree Units**: Digital tools for meetings, loan tracking, and attendance.
- **Service Providers**: Simplified request tracking and payment systems.

2.2 Assumptions and Dependencies

- Depends on government APIs for utility bills, health schemes, and insurance.
- Uses secure third-party APIs for face recognition, payments, and location services.
- Depends on verified data input by ward officials for accuracy.

3 System Features and Requirements

3.1 Functional Requirements

3.1.1 Kudumbashree Services

Use Case 1: Voice-to-Text and Summarization **Objective**: To convert meeting discussions into text and auto-generate meeting minutes.
Precondition: User must be attending a meeting session.
Postcondition: Meeting minutes generated and stored.
Flow of Events:

(a) User records meeting audio.

(b) System converts speech to text.

(c) System summarizes key points.

(d) Document stored in Kudumbashree records.

Use Case 2: Attendance (Face & Location) **Objective**: To mark attendance via face recognition and GPS.
Precondition: User must enable camera and location services.
Postcondition: Attendance recorded in the system.
Flow of Events:

(a) User opens app.

(b) Face scanned and verified.

(c) Location confirmed.

(d) Attendance marked.

Use Case 3: Loan Management **Objective**: To apply, approve, and track loans digitally.
Precondition: User must be a registered Kudumbashree member.
Postcondition: Loan status updated (approved/rejected).
Flow of Events:

(a) User submits loan request.

(b) Admin reviews and verifies.

(c) Loan decision updated in system.

Use Case 4: Organize Meetings **Objective**: To schedule and notify meeting events.
Precondition: User must be logged in.
Postcondition: Meeting created and members notified.
Flow of Events:

(a) Admin creates meeting event.

(b) System notifies members.

(c) Attendance and minutes recorded.

Use Case 5: Reports and Analytics Objective: To generate summaries of meetings, loans, and attendance.
Precondition: User must have access rights.
Postcondition: Reports generated.
Flow of Events:

(a) User requests report.

(b) System generates data summary.

(c) Report displayed or downloaded.

3.1.2 Waste Management

Use Case 6: Schedule Waste Pickup Objective: To request regular waste collection.
Precondition: User must be registered.
Postcondition: Pickup scheduled.
Flow of Events:

(a) User selects date and time.

(b) Request sent to waste collector.

(c) Pickup confirmed.

Use Case 7: Bulk Waste Pickup Objective: To handle bulk waste collection requests.
Precondition: User must specify waste type.
Postcondition: Bulk pickup scheduled.
Flow of Events:

(a) User submits request.

(b) Admin assigns vehicle.

(c) Collection carried out.

Use Case 8: Register Complaint Objective: To lodge complaints about waste collection.
Precondition: User must provide complaint details.
Postcondition: Complaint assigned to staff.
Flow of Events:

(a) User submits complaint with photo/location.

(b) System logs complaint.

(c) Admin assigns staff.

(d) Status updated until resolved.

Use Case 9: Reports and Analytics **Objective**: To generate reports on waste collection and complaints.
Precondition: Admin must be logged in.
Postcondition: Reports displayed.
Flow of Events:

(a) Admin requests report.

(b) System compiles data.

(c) Report generated.

3.1.3 Vehicle Services

Use Case 10: Owner/User Profile **Objective**: To manage vehicle owner and user profiles.
Precondition: User must be registered.
Postcondition: Profile updated.
Flow of Events:

(a) User enters profile details.

(b) System stores information.

Use Case 11: Book Vehicle **Objective**: To book a vehicle for travel.
Precondition: User logged in.
Postcondition: Vehicle booking confirmed.
Flow of Events:

(a) User requests vehicle.

(b) System finds available vehicle.

(c) Booking confirmed.

Use Case 12: Show Nearest Vehicles **Objective**: To display nearest available vehicles.
Precondition: User enables location.
Postcondition: Nearby vehicles displayed.
Flow of Events:

(a) User opens vehicle search.

(b) System fetches nearby vehicles.

(c) List displayed on map.

Use Case 13: Emergency Ride **Objective**: To request emergency vehicle instantly.
Precondition: User presses SOS.
Postcondition: Vehicle dispatched.
Flow of Events:

(a) User presses emergency button.

(b) Nearest vehicle identified.

(c) Ride confirmed.

3.1.4 Job and Education Services

Use Case 14: E-learning Chatbot **Objective**: To provide learning support using chatbot.
Precondition: User must be registered.
Postcondition: Response displayed.
Flow of Events:

(a) User asks question.

(b) System processes query.

(c) Chatbot provides response.

Use Case 15: CV Generator **Objective**: To generate CV automatically.
Precondition: User provides details.
Postcondition: CV generated.
Flow of Events:

(a) User inputs details.

(b) System formats CV.

(c) CV downloaded.

Use Case 16: Apply for Job **Objective**: To apply for listed jobs.
Precondition: User has a CV.
Postcondition: Application submitted.
Flow of Events:

(a) User selects job.

(b) Uploads CV.

(c) Application stored in system.

Use Case 17: User Dashboard **Objective**: To track job applications and training.
Precondition: User logged in.
Postcondition: Dashboard displayed.
Flow of Events:

(a) User logs in.

(b) System fetches applications and training data.

(c) Dashboard displayed.

3.1.5 Utility and Bill Services

Use Case 18: Pay Bills (Electricity, Water, Gas, Property Tax) **Objective**: To pay utility bills digitally.
Precondition: User must have a service account.
Postcondition: Bill paid and receipt generated.
Flow of Events:

(a) User selects bill type.

(b) Enters details.

(c) Payment processed.

(d) Receipt generated.

Use Case 19: Report Utility Issues **Objective**: To report issues in utilities.
Precondition: User submits complaint.
Postcondition: Ticket generated.
Flow of Events:

(a) User reports issue.

(b) System logs complaint.

(c) Staff assigned for resolution.

3.1.6 Civic Issues

Use Case 20: Report New Issues **Objective**: To allow citizens to report civic problems.
Precondition: User submits issue details.
Postcondition: Issue recorded.
Flow of Events:

(a) User enters issue details.

(b) System logs issue.

(c) Admin assigns staff.

Use Case 21: My Issues (Tracking) **Objective**: To track submitted civic issues.
Precondition: User logged in.
Postcondition: Issue status displayed.
Flow of Events:

(a) User requests status.

(b) System fetches data.

(c) Status shown.

3.1.7 Health Services

Use Case 22: Medicine Reminder **Objective**: To notify users about medicines.
Precondition: User sets schedule.
Postcondition: Reminder sent.
Flow of Events:

(a) User adds reminder.

(b) System stores schedule.

(c) Notification triggered.

Use Case 23: Health Record Vault **Objective**: To store personal health data securely.
Precondition: User uploads records.
Postcondition: Record saved.
Flow of Events:

(a) User uploads health record.

(b) System encrypts and stores record.

Use Case 24: Community Health Dashboard **Objective**: To provide overview of ward health statistics.
Precondition: Admin logged in.
Postcondition: Dashboard displayed.
Flow of Events:

(a) Admin requests dashboard.

(b) System compiles health data.

(c) Dashboard displayed.

Use Case 25: Blood/Organ Donation **Objective**: To connect donors with recipients.
Precondition: Donor registered.
Postcondition: Match found.
Flow of Events:

(a) Donor registers in system.

(b) Recipient request logged.

(c) System matches donor with recipient.

Use Case 26: Insurance and Health Schemes **Objective**: To provide access to government schemes.
Precondition: User enters details.
Postcondition: Eligible schemes displayed.
Flow of Events:

(a) User checks eligibility.

(b) System validates details.

(c) Schemes displayed.

3.1.8 Shop Services

Use Case 27: Stock Alerts **Objective**: To notify shopkeepers of low stock.
Precondition: Shop inventory linked.
Postcondition: Alert sent.
Flow of Events:

(a) Shopkeeper updates inventory.

(b) System checks stock.

(c) Alert sent if stock low.

Use Case 28: Local Product Catalog **Objective**: To display local shop products.
Precondition: Shopkeeper registers catalog.
Postcondition: Catalog displayed to users.
Flow of Events:

(a) Shopkeeper uploads products.

(b) System updates catalog.

(c) Citizens browse products.

Use Case 29: Home Delivery Request **Objective**: To request product delivery.
Precondition: User selects product.
Postcondition: Delivery scheduled.
Flow of Events:

(a) User places delivery request.

(b) Shop confirms.

(c) Delivery arranged.

Use Case 30: Feedback and Rating **Objective**: To collect citizen feedback on shops.
Precondition: User purchased product.
Postcondition: Feedback stored.
Flow of Events:

(a) User submits feedback.

(b) System records rating.

Use Case 31: Digital Payments and Billing **Objective**: To enable secure shop payments.
Precondition: User selects product.
Postcondition: Payment receipt generated.
Flow of Events:

(a) User makes payment.

(b) System processes transaction.

(c) Invoice generated.

3.2 Nonfunctional Requirements

- **Performance**: Must support at least 500 users concurrently in a ward.
- **Security**: End-to-end encryption for personal data, payments, and health records.
- **Usability**: User-friendly UI with multilingual support.
- **Scalability**: Expandable across multiple wards/districts.
- **Maintainability**: Modular design for easy feature updates.